

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 1 Part A — Uninformed Search Algorithms

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GitHub Repository Link:

<https://github.com/zaharaSHarouna/lab-3.git>_____

Date Submitted: ____29-07-2026_____

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	yes
Estimated percentage of the work influenced by AI	10%
Did you attach evidence of AI use where applicable?	yes

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Claude ai	I used it for a few debuggings	Why i am not seeing any graph(then I put the part of the code concerned)	Part 2 graph	I actually had just a typos in the variable name. The name didn't match so I wasn't working.
Claude ai	When I experimented with learning rate $lr=1.0$. It showed a warning and the program crash. So I have to solve it before to proceed	See the screenshot below	The learning rate experimentation part.	Myself know that the lr is too large, but I learned more when I asked AI the consequence of large lr , and how to safely pass it trough.

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft	See the screenshot below	Lab-3.ipynb
Prompt history	See the screenshot below	
Screenshot of AI interaction	See below	
Exported AI conversation	See below	
AI-generated code snippet	-	
Revised version showing student input	-	

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

```
/tmp/ipykernel_4028/890983563.py:14: RuntimeWarning: overflow
encountered in square_loss = float(np.mean((y_hat - y) ** 2))
/tmp/ipykernel_4028/890983563.py:18: RuntimeWarning: overflow
encountered in matmul_dh = d_yhat @ W2.T
/tmp/ipykernel_4028/890983563.py:19: RuntimeWarning: invalid value
encountered in multiply_dz1 = dh * (1 - np.tanh(z1)**2)
```

Jul 28

No worries! 🤖 This warning is expected and actually PROVES your point — $lr=1.0$ is too large and causes the gradients to **explode** (overflow)! The numbers get so large Python can't store them!

Let's add gradient clipping to handle the explosion gracefully:

Found the bug! 🐞 You created `ax` as an array but then used `ax1` and `ax2` which don't exist!

Note: `lr=1.0` caused a `RuntimeWarning: overflow` – the gradients became so large that floating point numbers couldn't store them! This is exploding gradients in action – exactly why gradient clipping is used in practice (especially in RNN/LSTM training).

Student Signature: ___Zahara Seybou

Harouna_____

Date: _____29-07-2026_____